

Magnetically Coupled Circuits

EEE203

Introduction

- The circuits we have considered so far may be regarded as conductively coupled, because one loop affects the neighboring loop through current conduction.
- When two loops **with or without contacts** between them affect each other through the **magnetic field generated by one of them**, they are said to be magnetically coupled.
- The transformer is an electrical device designed on the basis of the concept of magnetic coupling. It uses magnetically coupled coils to transfer energy from one circuit to another.

Self-Inductance

$$v = N \frac{d\phi}{dt}$$

Or,
$$v = N \frac{d\phi}{di} \frac{di}{dt}$$

Or,
$$v = L \frac{di}{dt}$$

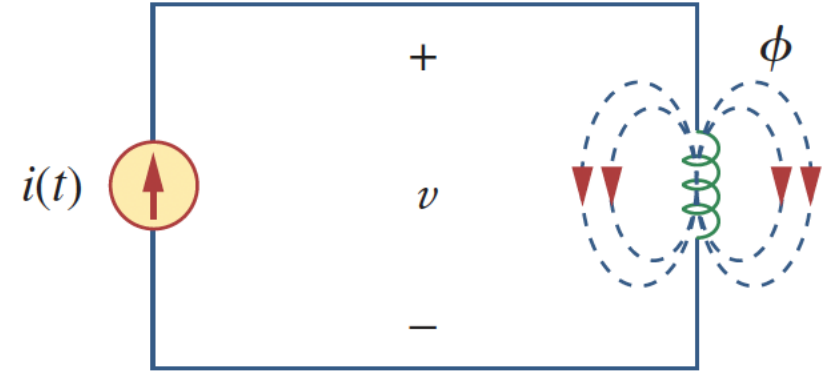


Figure 13.1

Magnetic flux produced by a single coil with N turns.

This inductance is commonly called **self-inductance**, because it relates the voltage induced in a coil by a time-varying current in the same coil.

$$L = N \frac{d\phi}{di}$$

Mutual Inductance

- When two inductors (or coils) are in a **close proximity** to each other, the magnetic flux caused by current in one coil links with the other coil, thereby inducing voltage in the latter. This phenomenon is known as mutual inductance.
- Mutual inductance is the ability of one inductor to induce a voltage across a neighboring inductor, measured in henrys (H).

Mutual Inductance

$\phi_1 = \phi_{11} + \phi_{12}$ One component links only coil 1, and another component links both coils.

$$v_1 = N_1 \frac{d\phi_1}{dt}$$

For coil 1

$$v_1 = N_1 \frac{d\phi_1}{di_1} \frac{di_1}{dt} = L_1 \frac{di_1}{dt}$$

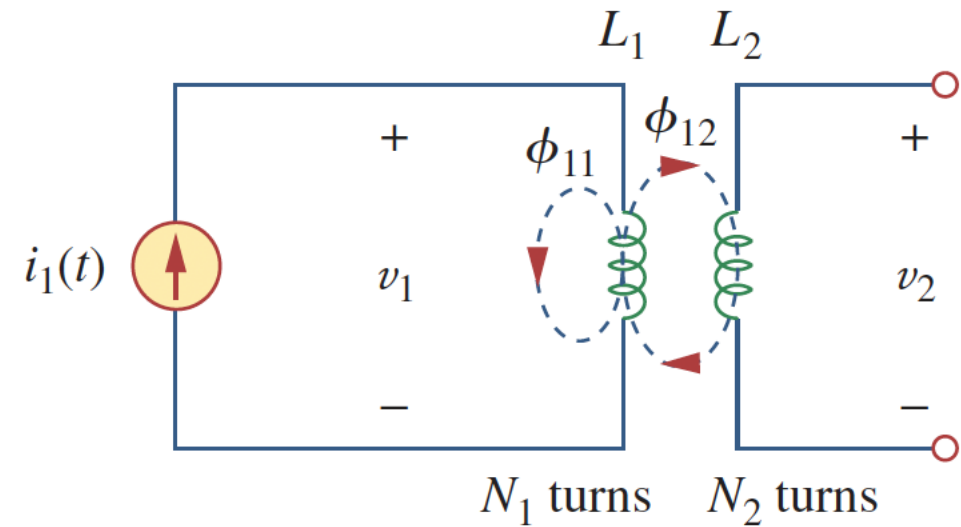


Figure 13.2

Mutual inductance M_{21} of coil 2 with respect to coil 1.

Mutual Inductance

$$v_2 = N_2 \frac{d\phi_{12}}{dt}$$

For coil 2 $v_2 = N_2 \frac{d\phi_{12}}{di_1} \frac{di_1}{dt} = M_{21} \frac{di_1}{dt}$

$$M_{21} = N_2 \frac{d\phi_{12}}{di_1}$$

M_{21} is known as the **mutual inductance** of coil 2 with respect to coil 1. Subscript 21 indicates that the inductance relates the voltage induced in coil 2 to the current in coil 1.

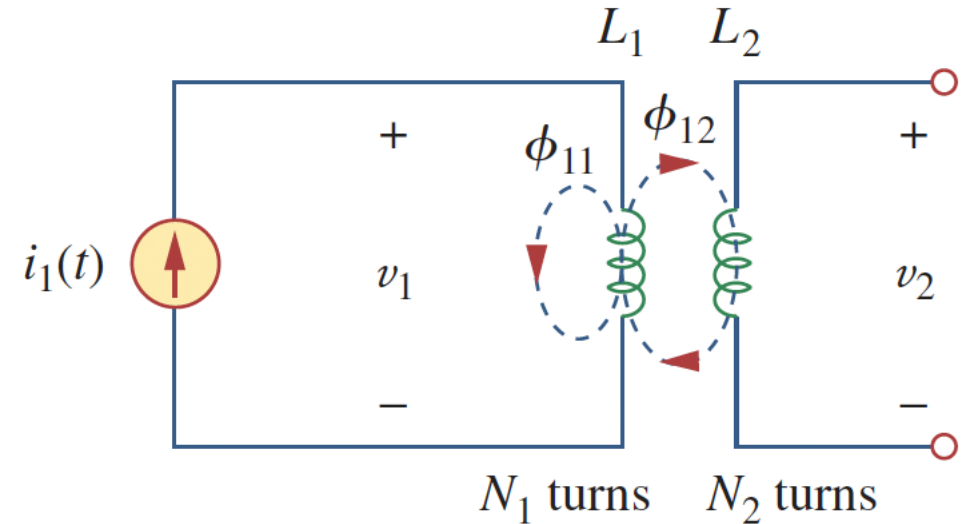


Figure 13.2

Mutual inductance M_{21} of coil 2 with respect to coil 1.

Mutual Inductance

For coil 2 $v_2 = N_2 \frac{d\phi_2}{dt} = N_2 \frac{d\phi_2}{di_2} \frac{di_2}{dt} = L_2 \frac{di_2}{dt}$

For coil 1 $v_1 = N_1 \frac{d\phi_{21}}{dt} = N_1 \frac{d\phi_{21}}{di_2} \frac{di_2}{dt} = M_{12} \frac{di_2}{dt}$

$$M_{12} = N_1 \frac{d\phi_{21}}{di_2}$$

M_{12} is the mutual inductance of coil 1 with respect to coil 2.

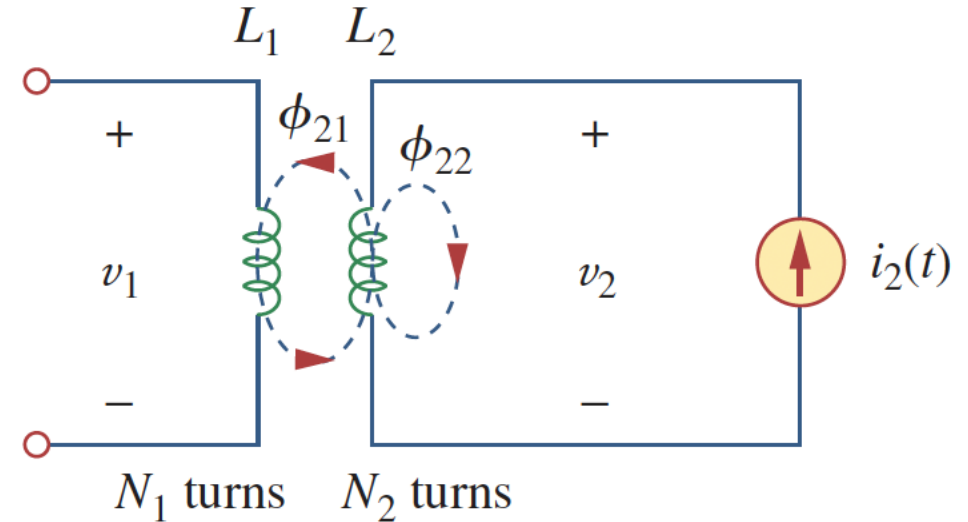


Figure 13.3

Mutual inductance M_{12} of coil 1 with respect to coil 2.

Mutual Inductance

- We can verify that:

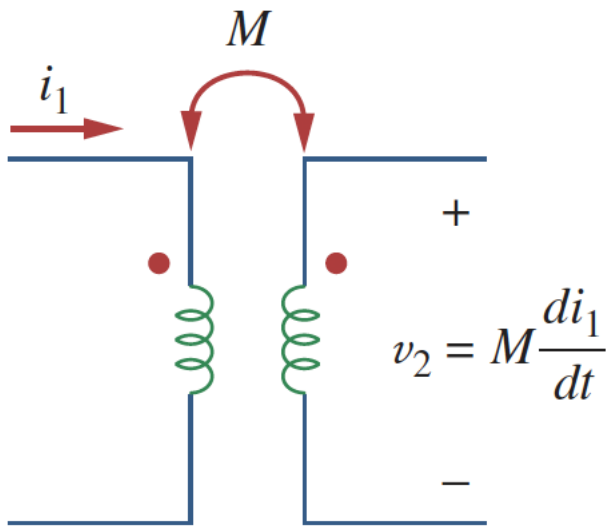
$$M_{12} = M_{21} = M$$

- Keep in mind that mutual coupling only exists when the inductors or coils are in **close proximity**, and the circuits are driven by **time-varying sources**. We recall that inductors act like short circuits to dc.

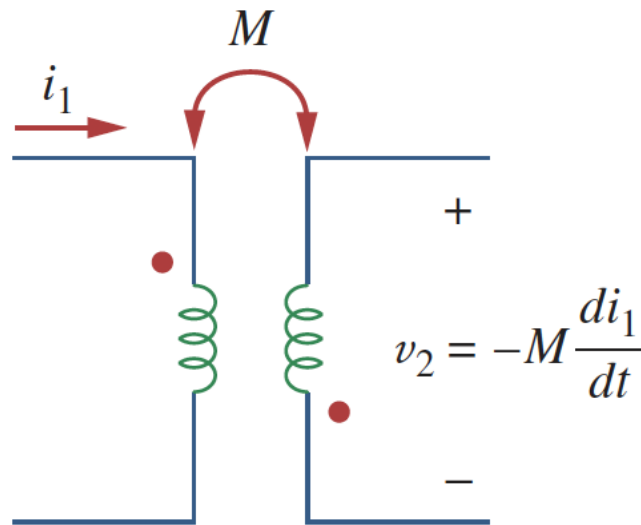
Dot Convention

- If a **current enters the dotted terminal** of one coil, the reference polarity of the mutual voltage in the second coil is **positive at the dotted terminal** of the second coil.
- If a **current leaves the dotted terminal** of one coil, the reference polarity of the mutual voltage in the second coil is **negative at the dotted terminal** of the second coil.

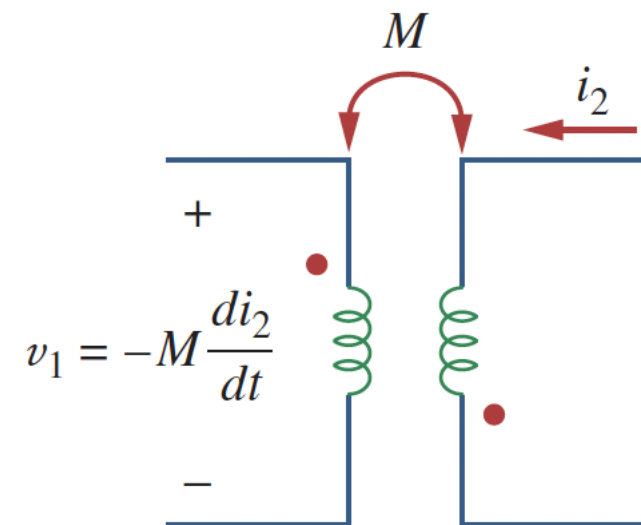
Dot Convention



(a)

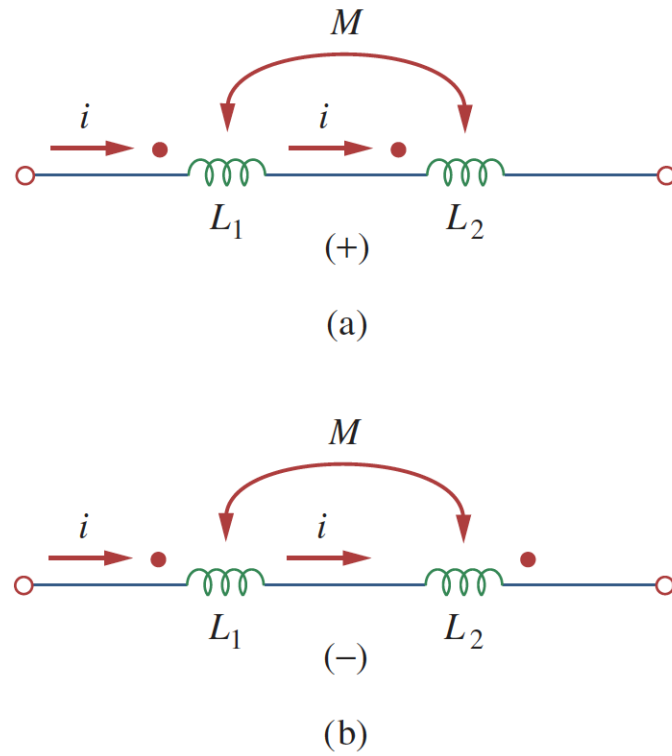
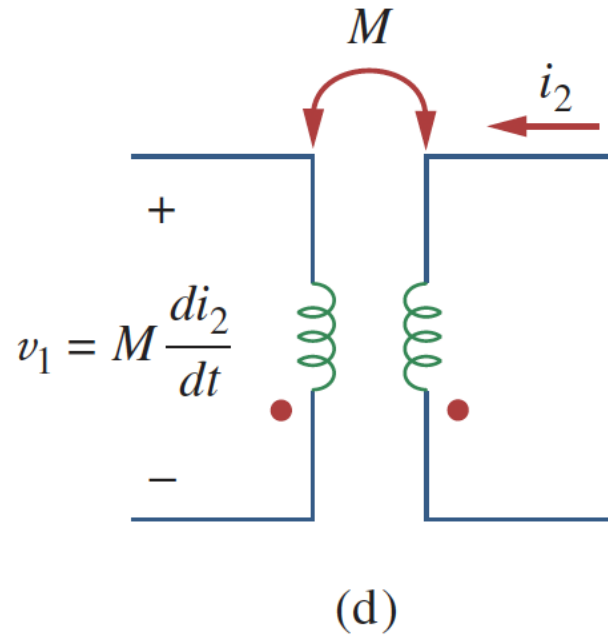


(b)

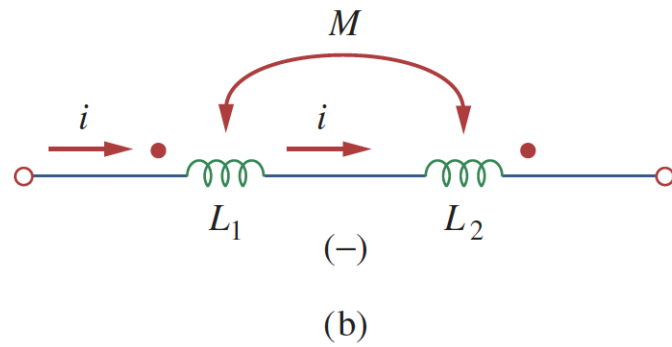


(c)

Dot Convention



$$L = L_1 + L_2 + 2M \quad (\text{Series-aiding connection})$$



$$L = L_1 + L_2 - 2M \quad (\text{Series-opposing connection})$$

Energy in a Coupled Circuit

**Instantaneous
energy
stored in the circuit**

$$w = \frac{1}{2}L_1i_1^2 + \frac{1}{2}L_2i_2^2 \pm Mi_1i_2$$

The positive sign is selected for the mutual term if both currents enter or leave the dotted terminals of the coils; the negative sign is selected otherwise.

**Coefficient
of coupling k**

$$k = \frac{M}{\sqrt{L_1L_2}}$$

The coupling coefficient k is a measure of the magnetic coupling between two coils; $0 \leq k \leq 1$.

If the entire flux produced by one coil links another coil, then $k=1$ and we have 100 percent coupling, or the coils are said to be perfectly coupled. For $k < 0.5$, coils are said to be loosely coupled; and for $k > 0.5$, they are said to be tightly coupled.

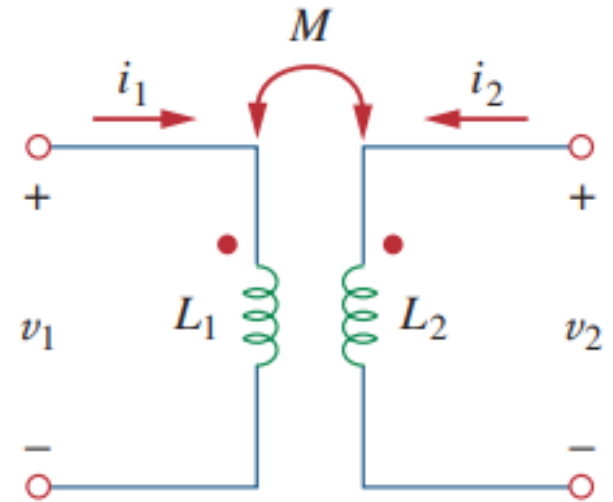


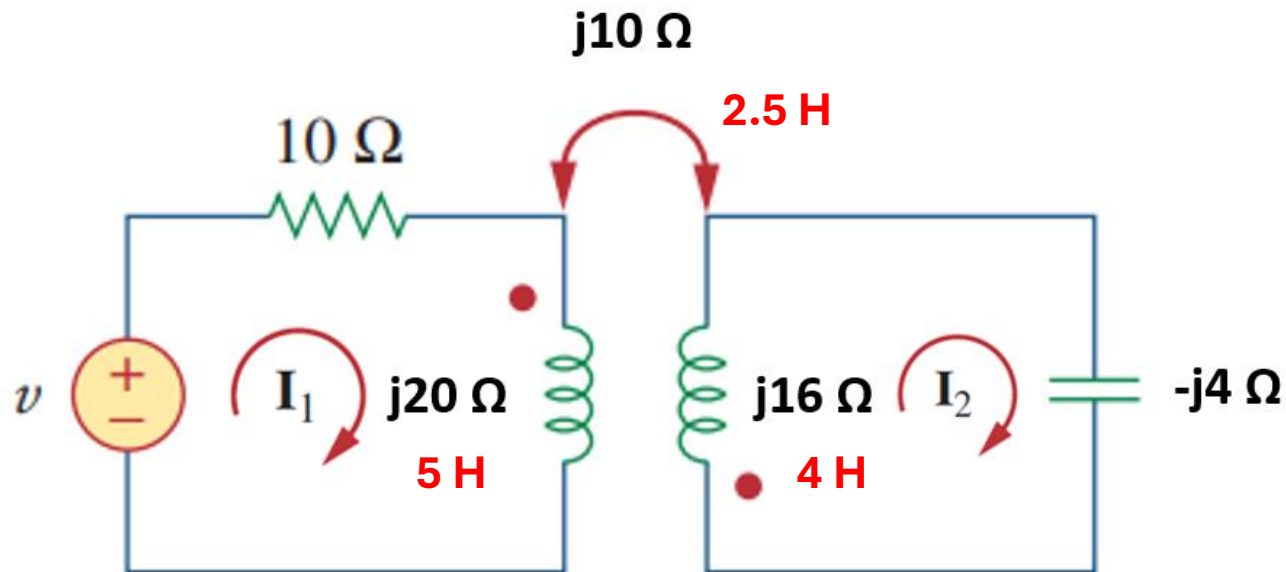
Figure 13.14

The circuit for deriving energy stored in a coupled circuit.

Example #1

Let's $v = 60 \cos(4t + 30) \text{ V}$.

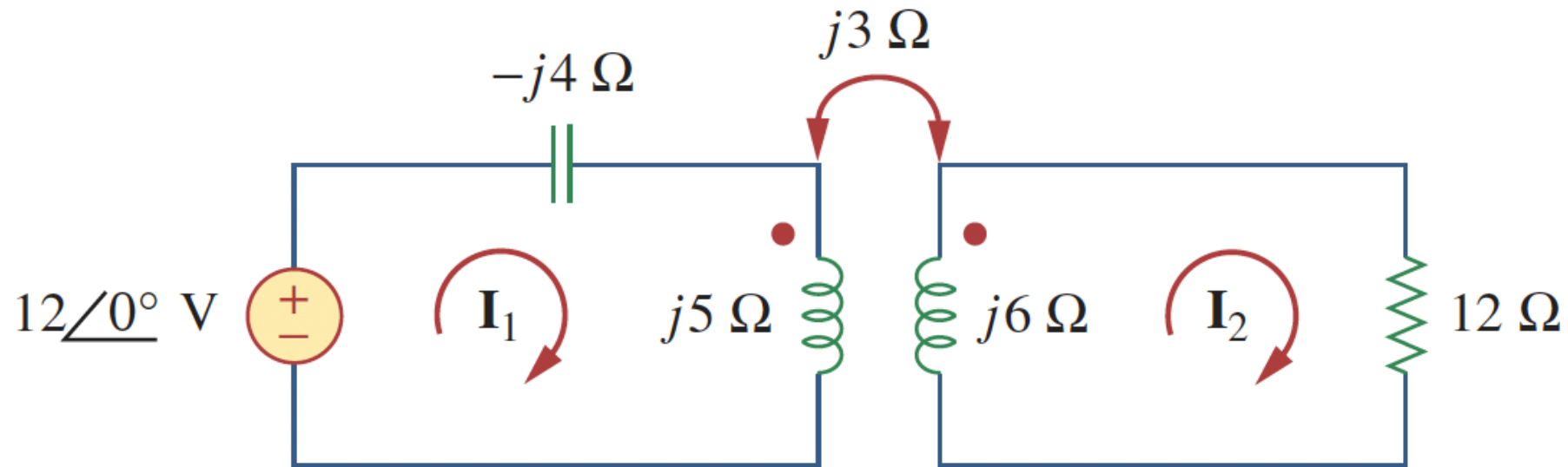
- Determine the coupling coefficient.
- Calculate phasor currents I_1 and I_2 in the circuit
- Calculate the energy stored in the coupled inductors at time $t=1\text{sec}$.



Circuit for Example #1

Example #2

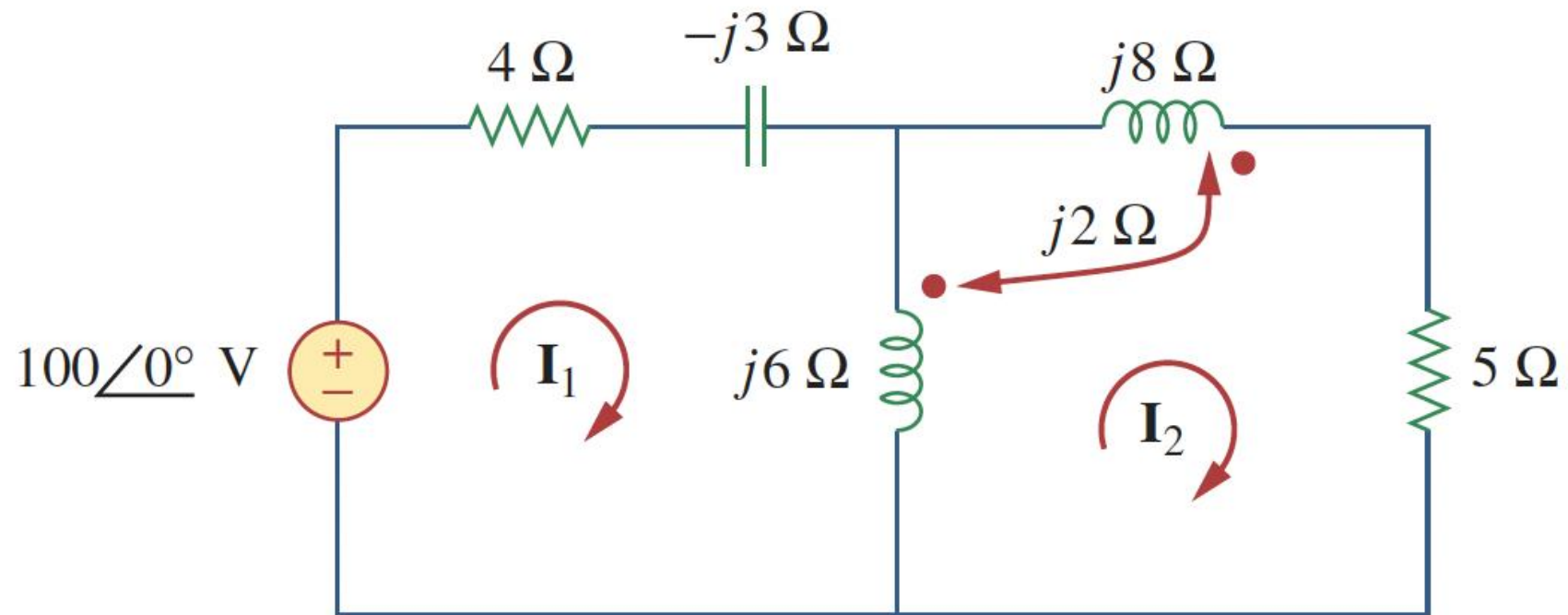
Calculate the phasor currents I_1 and I_2 in the circuit.



Circuit for Example #2

Example #3

Calculate the mesh currents in the circuit.



Circuit for Example #3